Review Article

Advances in Polymer Synthesis and Properties: Current Trends and Applications

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ABSTRACT

Polymer is the term for a large group of compounds made up of many smaller molecules. Almost every industry uses polymers. Plastics, cotton, rubber, wool, and Teflon are all types of polymers. The fact that cellulose's infrared absorption spectra are very similar to those of a spectrum that comes from space is proof that cellulose really does exist. The cellulose pellicle is the outer layer of the material. Most elastic materials, like the metals used in springs, show elastic behaviour because their bonds are stretched out. When rubber is relaxed, it has many places where long polymer chains connect to each other in a coiled shape. The main goal of polymeric delivery systems is to deliver drugs in a way that is either controlled or continuous. Polysaccharides have also been used to make drugs more concentrated in the colon after they are taken orally. Polymeric materials could lead to many interesting new uses in the not-too-distant future. A lot of different businesses are finding new uses for polymers.

Introduction

Polymer is derived from the Greek words "poly," meaning "many," and "mero," meaning "parts." Molar masses ranging from dozens to millions indicate the immense size of polymers. In the organic chemical sector, almost 80% of the production is devoted to synthetic polymers. These include plastics, textile fibers, and synthetic rubbers [1]. The large class of materials known as macromolecules includes polymers, which are made up of numerous smaller molecules known as monomers, wthathich can be connected to form lengthy chains depicted in Figure 1. Polymers' adaptability has allowed their use in the production of tars, gums, resins, oils, and tar products for millennia. Regardless, it was during the industrial revolution that the contemporary polymer sector started to take form. Using a process known as "vulcanization," Charles Goodyear developed a usable form of natural rubber in the late 1830s [2,3]. To start, glycosidation joins cellulose residues into the (1-4)-shape that makes cellulose a naturally occurring biological photopolymer [4]. Bundles of glucan chains are connected by hydrogen

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bonds to create the microfibril, which is a very tiny, insoluble thread. Given that multicellular organisms at this evolutionary stage can also produce it, it's reasonable to assume that mammals, including humans, could potentially have this capability as well. These creatures are known as chordates [5]. One of them is cellulose, a naturally occurring biological photopolymer composed of (1-4)-shaped cellulose residues connected by glycosidation. Microfibrils are insoluble, extremely tiny threads composed of hydrogen-bonded bundles of glucan chains. Mammals, including humans, may be able to synthesize cellulose in the future, given that multicellular creatures in this evolutionary stage are capable of doing so. Every chordate is before conducting polymers, also called conjugated polymers, were discovered; polymers were thought of as electrical insulators. Polymers are now widely recognized as organic molecules that possess exceptional electrical and optical properties, comparable to those of inorganic semiconductors [6-8]. The conjugated carbon chain's highly delocalized, polarized, and electron-dense π bonds impart their electrical and optical properties. The chain exhibits a complicated structure that is typical of conducting polymers; it alternates between single and double bonds. Polyacetylene (PA), polyaniline (PANI), polypyrene (PPy), polythiophene (PTH), poly(*p*-phenylene) (PPP), poly(*p*-phenylenevinylene) (PPV), and polyfuran (PF) are all part of this broad class [8]. Doping sulfur nitride metal, an inorganic material, with bromine improved its electrical conductivity, according to MacDiarmid and physicist Alan J. Heeger. Because of this finding, conducting PA was investigated further [9]. Research showing that bromine-doped PA had a conductivity one million times higher than pure PA was awarded the Nobel Prize in Chemistry in 2000. There could be thousands, if not millions, of monomer units that comprise traditional polymers. The polymers are nevertheless rigid and solvent-soluble, even if conjugated polymer chains have fewer monomer units. Its mechanical property is a result of its composition, which alternates between single and double bonds. While the processability and solubility of conducting polymers are largely dictated by the connected side chains, the mechanical, electrical, and optical properties are conferred by the dopant ions attached to them. Conducting polymers can be either fully crystalline or slightly amorphous [10]. A phase change transforms an insulator into a metal with electrically charged particles like solitons, polarons, and bipolarons, this happens when π-bond delocalization occurs in conducting polymers due to disorder [11]. Although pure conjugated polymers are electrical insulators, their conductivity grows in relation to the dopants added. Their mechanical and electrical characteristics are similar to those of common polymers in their undoped state, and they behave similarly to conventional semiconductors in that they are anisotropic and have a quasi-one-dimensional electronic structure with a moderate bandgap of 2 eV. When they are dopped, or upon photoexcitation the structure of polymers can alter, leading to a self-localization of the p bond and subsequent nonlinear excitation of various types (*e.g.,* polarons, solitons, bipolarons, etc.), ultimately leading to a metallic state [12-14]. In this comprehensive review, every aspect of conducting polymers are covered, from their various applications and properties to their production techniques (Figure 1, Table 1).

Chemical-Property

Most of the time, the intermolecular attractive forces determine a polymer's properties. Because polymer chains are so long, the forces between them are much stronger than the forces between regular molecules [15]. Dipoles in the monomer units might change how molecules in polymers interact with each other. Figure 2 shows how amide or carbonyl group-containing polymers can form hydrogen bonds between chains that are next to each other. A PEGylated lipid bilayer-supported mesoporous silica nanoparticle (MSN) composite for dual drug synergistic co-delivery is engineered with the C=O groups on one chain strongly attracted to the partially positively charged N-H groups on the other [15,16]. The process of turning a monomer into a polymer Monomers are small molecules that make up larger molecules called polymers. Most of these molecules are organic [17]. A single monomer molecule can form

chemical bonds with two or more other monomers. Polymers are substances that people create by putting together many monomers. Polymers consists of networks of monomeric units that are not well understood. Polymerization is a chemical reaction that occurs when several monomer units combine to form a polymer. Polymerization can produce macromolecular structures that are either linear or branched. Another possible shape is a complex three-dimensional network. The diagram below shows the basic steps for turning monomers into polymers [18].

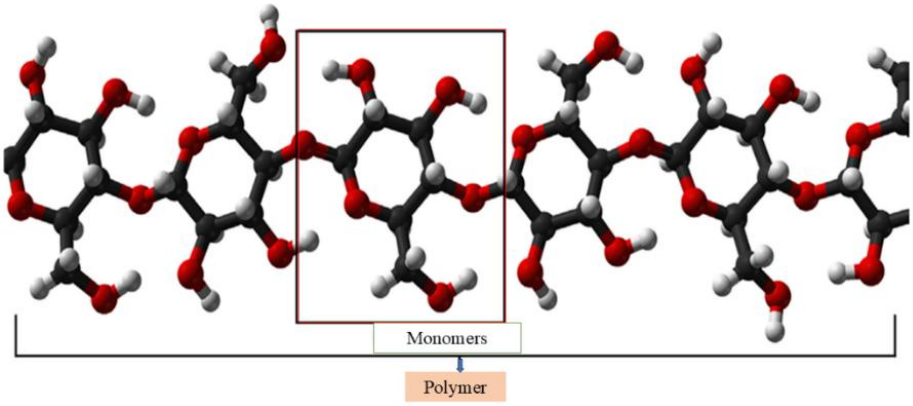


Figure 1: General structure of the polymer

Table 1: Types of polymers and their chemical properties

Polymer type / Class	Structural / Chemical features	Key chemical Properties / Behaviour	Examples
Homopolymers	Single kind of repeating monomer unit throughout chain	Uniform composition; predictable crystallinity, melting point, and mechanical properties	Polyethylene (PE) and polystyrene (PS)
Copolymers	Two or more types of monomer units in chain	Properties can be tuned (*e.g.,* block, random, alternating arrangement	acrylonitrile-butadiene-styrene (ABS), SBR, styrene/acrylonitrile, *etc*.
Thermoplastics	Mostly linear or lightly branched chains, minimal crosslinking	Can soften / melt on heating and re-solidify on cooling (reversible). Soluble or swellable in certain solvents Generally lower chemical resistance (depends on backbone)	PE, Polypropylene (PP), Polyvinyl chloride (PVC), and PS
Thermosets	Highly crosslinked three-dimensional networks	Become irreversibly set upon curing; do not melt on reheating; good thermal stability, chemical resistance, mechanical rigidity	Epoxy resins, phenolic resins, melamine formaldehyde, and polyester (crosslinked)
Elastomers (Rubbers)	Flexible, lightly crosslinked, often amorphous below Tg	Very high elongation, elastic recovery, low modulus, good resilience, often low Tg	Natural rubber, styrene-butadiene (SBR), and polyurethane elastomers

Linear polymers	Straight chains without branching or crosslinks	Higher density and crystallinity (if able to pack), higher melting points, stronger intermolecular forces	High density polyethylene (HDPE), some nylons, and linear polyesters
Branched polymers	Chains with side branches off main backbone	Less efficient packing, lower density, lower melting point, more amorphous regions	Low density polyethylene (LDPE), glycogen, amylopectin etc.
Crosslinked / Network polymers	Chains interconnected by covalent bonds in 3D network	Rigid, often insoluble, high thermal and chemical resistance, brittle	Bakelite (phenol-formaldehyde), urea formaldehyde, melamine formaldehyde
Amorphous polymers	Lack of long-range ordering / crystallinity	Transparent (often), glass transition temperature (Tg) dominates behavior, lower density, and higher free volume	PS, polymethyl methacrylate (PMMA)
Crystalline (semi-crystalline) polymers	Regions of regular chain packing (crystallites) plus amorphous domains	Higher density, higher melting point (Tm), better stiffness, lower permeability, and opacity	PE, PP, and nylon (when crystallizable)
Biodegradable / Biopolymer types	Polymers designed to degrade (biological or chemical)	Break down via hydrolysis, enzymatic attack, etc.; lower long-term stability; varying mechanical properties	polylactic acid (PLA), polyhydroxyalkanoates
Conductive / Functional polymers	Include conjugated bonds or dopants; special functional groups	Exhibit electrical conductivity, semiconducting behavior, responsive to stimuli	PANI, polypyrrole, and PA

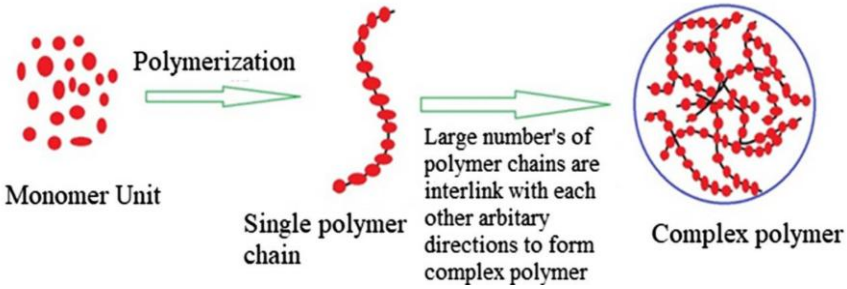


Figure 2: Monomer to polymer conversion

Mechanical Characteristics

When it comes to how they will be used, the bulk properties of most polymers are what really matter. These properties determine how the polymer behaves on a larger scale. The tensile strength of a material is the amount of stress it can handle before it permanently changes shape [14,19].

Property of Transport

Transport properties like diffusivity affect how quickly molecules can move through a polymer matrix. In many membrane and film applications, polymers depend on them greatly. When discussing polymers, "melting point" is a better term than "solid-liquid phase transition" to describe the change from a solid to an amorphous solid [20].

Elasticity

Most elastic materials, like spring metals, act elastically because their bonds are distorted. When bond lengths change from their (least energy) equilibrium due to force, strain energy is stored electrostatically (Figure 3) [21].

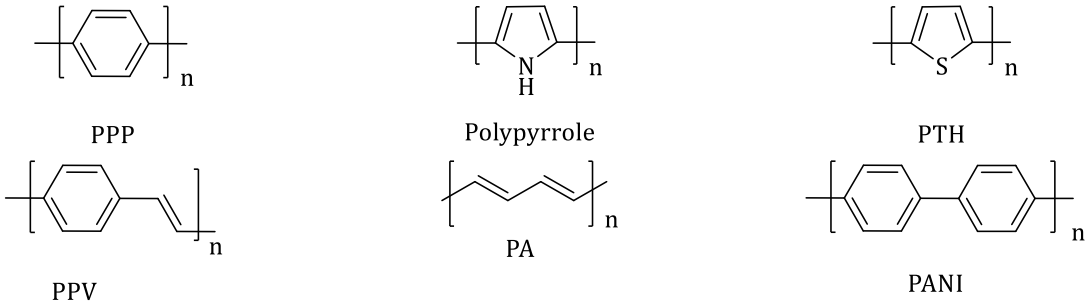


Figure 3: Polymer conversion [22]

Classification

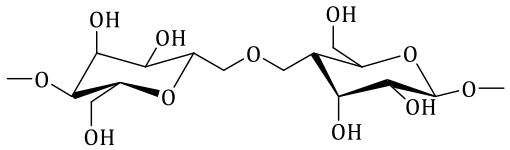
Natural polymers come from plants and animals. These polymers are what keep us alive [17-23]. Here they are:

(a) Starch: A polymer of glucose that serves as a plant's glycogen reserve. Plants are made of the polymer cellulose. Paul used this strong material, along with wood and stems, to build his treehouse. Starch from maize (Figure 4). Another option is woven fabrics made from hemp and cotton, which are both made of cellulose. Starch is also made by a number of plant species. There is too much grain, corn, and potato starch [24].

(b) Cellulose: Cellulose is another crucial component, just like glucose polymers. It is the most important part of a plant's structure. Photosynthesis produces glucose, which is a step in the synthesis cellulose and starch. Cellulose is a polysaccharide composed of a long chain of *D*-glucose units that are linked by hundreds to thousands of β (1→4) bonds. The formula for this organic molecule is $(C_6H_{10}O_5)_n$. Cellulosic ethanol, produced from cellulose by certain bacteria, is one type of ethanol that works well as a fuel. Figure 4 shows cellulose [25].



A- Starch in Corn



B- Cellulose

Figure 4: Natural polymers

(c) Proteins: These polymers of α-amino acids are usually made up of 20 to 1,000 α-amino acids in a very specific way. They are an important part of the diet because they are the building blocks of animals [26].
(d) Nucleic acids: These polymers differ from one another because they are composed of nucleotides. RNA and DNA are two common types of nucleic acids. Biopolymers, such as proteins, nucleic acids, and polysaccharides (like starch, cellulose, and others), play a significant role in controlling much of plant and animal life. Nucleotide chains, which are linear polymers, serve as the building blocks of nucleic acids. There are three parts to a nucleotide: the pentose sugar, the phosphate group, and the nucleobase, which can be either pyrimidine or purine [27].
(e) Silk: Silk, which some caterpillars produce, is another excellent source of protein. For many generations, people have created beautiful clothes out from this fabric. Additionally, nylon was developed to resemble silk, but it still does not compare to the real thing [28].
(f) Enzymes: Enzymes are a type of protein that performs specific functions in the body. Enzymes are small, specific protein clusters that accelerate certain biological processes. These processes would be too slow to sustain life without enzymes. An unexpected fact is that enzymes can produce their own type of enzyme [29].
(g) Fibers: Fibers have a significant appplication in polymeric materials. They can be made from many different subtances, including plastics and elastomers [30].

Natural Polymers and their USES in Medicine

There are many biocompatible materials on the market, such as natural hydrogels, biodegradable scaffolds for tissue regeneration, biomimetic polymers, and sustained release systems [31,32].
Using only natural polymers, this is a creative way to deliver drugs to the colon.
A new type of colon-targeted tablet was made using pectin as a carrier. The model medications were Indomethacin and Diltiazem HCl. After being covered in inulin and shellac, the tablets were tested for hardness, coat thickness, and average weight. For six hours, a colonic fluid simulator was used; for three hours, a pH 7.4 buffer was used; and for two hours, a pH 1.2 HCl buffer was used to examine how the finished tablets released *in vitro* [33].

Using Cellulose

The main goal of polymeric delivery systems is to control or maintain medicine in a steady state. Polysaccharide-based hydrophilic matrices are a common biomaterial for dosage forms that release drugs over time. This is due to cellulose being the most common biopolymer that occurs naturally. Hydroxypropyl methylcellulose, microcrystalline cellulose, hydroxyethyl cellulose, and hydroxypropyl cellulose can all be used to make time-controlled delivery systems. Microcrystalline cellulose, hydroxyethyl cellulose, hydroxypropyl methylcellulose, sodium carboxymethyl cellulose, and hydroxypropyl cellulose are also in the coating of the tablet [34].

Giving through the Nose

The nasal mucosa is a perfect place for bioadhesive medicine delivery systems to function. Chitosan drug delivery systems, such as gels, liposomes, and microspheres, are ideal for expansion on the nasal mucosa due to their robust bioadhesive characteristics [35,36].
Chitosan is a great polymer to use for buccal administration because it adheres to mucous membranes and improves absorption. Three hours after pharmacological administration via the rabbit sublingual route, directly compressible bioadhesive tablets with a weight ratio of 1:4 of ketoprofen, chitosan, and sodium alginate exhibited sustained release [37].

Delivery of Drugs through the Digestive System

The density of floating systems is lower than that of gastric fluid. As a result, medications administered through the upper gastrointestinal tract exhibit enhanced bioavailability due to their extended gastric retention time [38]. Oral administration is an option for some medications, such as insulin for diabetes and 5-

fluorouracil for colon cancer, instead of giving them through an IV [39].

Delivery to the Colon

Chitosan has made it so that medicines taken by mouth now last longer in the body. Recent studies have demonstrated that bacteria in the colon can degrade chitosan [40].

Delivery through the Vagina

Chitosan has made oral drugs last much longer in the body. Recent studies have demonstrated that bacteria in the colon can degrade chitosan [41].

Delivery through the Skin

Chitosan's ability to produce films is excellent. The devices can release medicine based on how strong the film's cross-linking is and how thick the membrane is. Microspheres and beads that contain an *in situ* synthesized chitosan-alginate polyelectrolyte complex (PEC) present innovative prospects in controlled release systems, wound dressings, and packaging [42].

Biopolymer and the Categorization of Biopolymers

Biopolymers are natural polymers that are found in living things. There are many types of living things, include plants, animals, microbes, and more. Biopolymers are natural polymers that are found in living organisms. These include plants, animals, microorganisms, and more [43].
Natural polymer
The "natural" type of biopolymer comes from living organisms and is one type of biopolymer. Biopolymers such as proteins and polysaccharides originate from nature (Figure 5) [44].

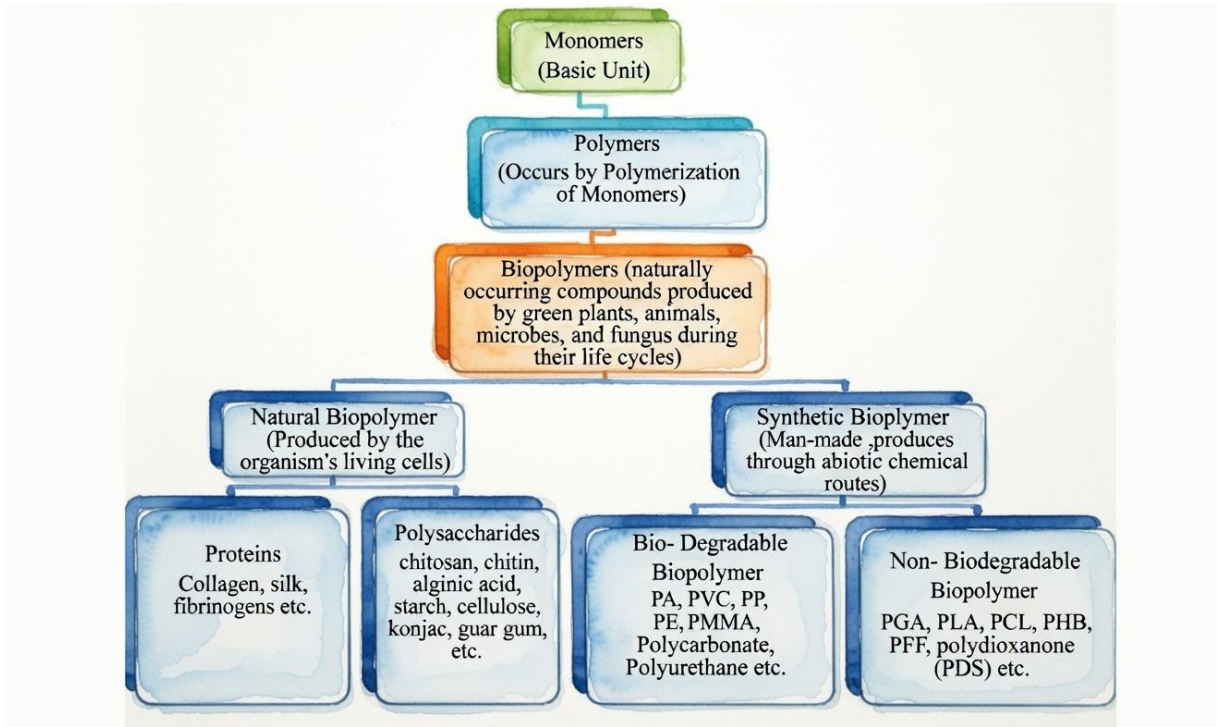


Figure 5: Classification of biopolymers [45]

Workshop Processes

Using Guar Gum in Medicine Using Guar Gum

Pharmaceutical companies use guar gum powder for many different purposes, such as thickening, stabilizing, emulsifying, binding, controlling water retention or phase-out,

suspending, pour control, and process aid [37,46-48].

Guar Gum for Beauty

This colloid is in lotions and creams for skin care. It thickens and keeps the mixture fresh. It also helps dispense toothpaste and shaving cream out from their tubes. Guar gum powder for industrial use that is of commercial grade Powdered guar gum can be used in wastewater treatment as a thickener, emulsifier, binder, wet-end strength additive, gelling agent, water barrier, flocculation aid, sizing agent, and many other useful applications [49].

Guar Gum for the Paper

Guar gum has everything that other solutions do not have. It makes the surface of the printing paper thicker [50].

Guar Gum for the Fabric

Because it makes great films and thickens substances, guar gum is used in textile finishing, printing, and sizing. The sizing process works better, and there is less warp breakage and dusting as a result [51].

Guar Gum in Oil Field Uses

Guar gum powder is produced for industrial use and is employed in a lot of different ways to stabilize, thicken, and hold solids in place. These methods include oil well stimulation, mud drilling, and oil well fracturing. Because it is slurrible and contains dispersible natural guar gum in it, diesel hydrates quickly. Guar gum is a synthetic polymer, defoamer, and surfactant that meets all of the rheological needs of water- or brine-based drilling fluids. This is why it is so widely used in the oil field [52].

Fibers

Fibers are made up of a wide variety of polymeric materials, such as plastics and elastomers. They have a significant impact on many areas. People have been using natural fibers for a long time. Some examples are silk, wool, and cotton. The invention of synthetic silk in 1885 marked the start of a new era in fiber production [53].

Textile Use

Rubber, also known as elastic, is a fiber that is very important in the textile industry because it can stretch and return to its original shape. One kind of rubber fiber is rectangular rubber fiber, which is made by cutting extruded film into strips. There is also the extruded round type, which can be used for many things [54].

Growth of the Body

"You are what you eat," as the saying goes. Protein is a natural polymer that makes up the bodies. It is consumed all the time [55].

Using Rubber

Rubber is a very flexible material that is used in a wide range of industrial and consumer goods, from finished products to intermediate goods. Rubber is often used to make tires and inner tubes. The remaning 44% of the market is made up of general rubber goods (GRG), which are not tires or tubes [56].

Art and Painting

Rubber is used in a wide range of industrial and consumer goods, from finished goods to intermediates, because it is so adaptable. People often use rubber to make tires and inner tubes. The GRG business makes up the other 44% of the total [57].

Making Paper

The papermaking industry uses more starch than any other non-food industry on Earth, over 1 million metric tons a year. For example, copy paper can have up to 8% starch in it. There are both chemically modified starches and natural starches used in making paper [58].

Adhesives for Corrugated Board

The papermaking industry uses the most starch each year of all non-food industries, millions of metric tonnes. For example, copy paper can have starch levels as high as 8%. Starches that are both natural and chemically modifed are used to make paper [59].

Starch for Clothes

Clothes or cleaning the house to make starch, you need to heat water and vegetable starch. In the past, you also had to boil the mixture. In the 16th and 17th centuries, Europeans often used starch to make the delicate linen collars and ruffs that well-off people wore around their necks stiffer [60].

Making Conductive Polymers

PA plastic- The Nobel Prize was rewarded to the individuals who worked on improving PA's conductivity by adding other materials to it. PA and its derivatives have many uses because of their excellent properties. If you look more closely, you can see that there is a lot to learn about it, such as its electrical conductivity, photoconductivity, liquid crystal properties, and chiral identification. A linear polyene chain comprises the main chain of PA. PA is flexible and can be decorated with pendant groups, so you can create monosubstituted or disubstituted PA by swapping out hydrogen molecules for foreign molecules at the other carbon. Some of the ways to make conducting polymers include electrochemical polymerization, chemical oxidation, vapor phase synthesis, hydrothermal, solvothermal, electrospinning, self-assembly, photochemical, inclusion, solid state, and so on [61].

$$HC{\equiv}CR \rightarrow [-(HC{=}C(R))-]_n$$

Monosubstituted Polyacetylene

$$H{-}C{\equiv}C{-}R' \rightarrow [-(R)C{=}C(R')-]_n$$

Disubstituted Polyacetylene

There are many ways to make conductive polymers, such as electrochemical polymerization, vapor phase synthesis, electrospinning, hydrothermal, solvothermal, self-assembly, photochemical, inclusion, solid state, and plasma polymerization. You can improve the electrical conductivity and optical properties of conducting polymers by adding the right components to them. When the doping level goes up, the conductivity of PA, which is usually around 10^{-5} s cm^{-1}, increases significantly to 10^2 to 10^3 s cm^{-1}. You can change the conductivity to have different properties, such as electrochemical or optical-mechanical ones, depending on the doping substance [62].

Polymers that Can Conduct Electricity and Be Used in Supercapacitors

The demand for fossil fuels like coal, natural gas, and others is greatly affected by the state of the world economy. The rapidly declining availability of fossil fuels is aggravating numerous social and environmental issues. Climate change, global warming, and health all mean that are needed to fix problems with current energy sources. This need has led to research into renewable energy sources that are both long-lasting and high-energy, such as supercapacitors, fuel cells, and wind energy. Businesses that make supercapacitors are interested in the future of wearable technology, electric cars, and other related fields. A supercapacitor can store a thousand times more energy than a regular capacitor. They also have a long cycle life, a high-power density, and a fast charge-discharge cycles [63,64].

There are three main types of supercapacitors based on how they store charge. An EDLC's non-Faradic process makes charge build up at the interface between the electrode and the electrolyte. In an electrochemical system, the capacitance is the result of two charge layers. Because this energy storage system is mostly in charge of controlling electrophoretic liquid chromatography (EDLC), carbon nanotubes, graphene, and other carbon-based materials with large surface areas are often used as electrodes. Unlike redox reactions or

intercalation processes, pseudo capacitors are powered by faradic processes [64,65] and are responsible for storing charges. Pseudo capacitors have more power and energy than EDLCs. Conducting polymers and transition metal oxides, like Mn_3O_4 or Co_3O_4, are common materials for the electrodes in pseudo capacitors because they can undergo redox reactions. An EDLC/hybrid supercapacitor combines the best features of both types of capacitors [66]. EDLCs have the right amount of specific capacitance and cycle stability, but pseudo capacitors have a lot of power and energy density. A hybrid capacitor takes the best parts of both an EDLC and a pseudo capacitor [67]. Conducting polymers are useful in many fields because they have good electrical properties, high redox activity, good electrochemical behaviour, strong wave absorption, and high theoretical capacitance [68]. There are many different shapes that conducting polymers can take, such as sheets, particles, hydrogels, rod-like structures, and more. These shapes show how these polymers behave in terms of light, electricity, and mechanics. To fix the cycle stability problem with PANI, it is usually mixed with carbon-based materials like graphene, activated carbon, or graphite [69]. PANI-based carbonaceous composites have one major problem: they clump together during synthesis. This reduces the performance of supercapacitors because there are fewer active sites and more ion mobility. Experts focused on three-dimensional porous carbons as a potential solution to this problem. In this case, ammonium persulfate was used in an *in situ* chemical oxidative manner to make the composite [70].

Conclusion

Humanity will utilize polymeric materials in the same way that nature has utilized biological polymers. Synthetic polymers have been the material of choice for people ever since the Stone Age, and they will continue to be so through the Bronze Age, the Iron Age, the Steel Age, and even into the Modern Age of Polymers. The fact that there are so many applications for polymeric materials, both now and in the future, has resulted in a significant increase in the demand for highly skilled individuals to conduct research and development in the field of polymer science and engineering. People who choose to work in this industry may discover that they have a job that is both financially secure and meaningful.

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